SPE-229550-MS

An Analytical Approach to Proppant Transport Mechanism for Numerical Simulation in Unconventional Reservoir

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This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

The "horizontal well + hydraulic fracturing" approach has become a key development method for low-permeability oil and gas resources. However, the impact of natural fracture development and fracture opening on proppant transport and placement in the primary hydraulic fracture remains unclear. This study employed a Euler–Euler proppant transport model to investigate the effects of geological factors (fracture aperture, fracture count) and engineering factors (injection rate, fracturing fluid viscosity, sand ratio, and proppant size) on proppant migration and placement within complex fracture networks. Results indicate that: (1) fracture opening significantly reduces the height of stable sandbanks in the primary hydraulic fracture, with maximum reductions of 24.67% in characteristic length and 50.37% in characteristic area compared to a single vertical fracture; (2) fracture count mainly affects the height from the fracture base to the top of the stable sandbank (HLB); (3) higher injection rate, sand ratio, and proppant size shorten the time to reach HLB, whereas larger fracture aperture, fracture count, and fluid viscosity extend it; under bedding-plane opening conditions, the maximum reductions reach 34.67% in length and 49.06% in area; (4) in narrow fractures, proppant forms a stable sandbank near the fracture entrance, while in other cases it remains suspended. These findings provide a theoretical basis for the design of low-permeability and unconventional reservoir development.

Introduction

Currently, the "horizontal well + hydraulic fracturing" technology is widely applied in the development of low-permeability oil and gas reservoirs, and is commonly adopted in pilot development projects in countries such as China, the United States, and Canada. This technology plays an important role in enhancing recovery and productivity. Studies have shown that in low-permeability reservoirs, the presence of bedding weak planes has a significant impact on fracture creation, with bedding planes controlling the propagation of hydraulic fractures. Compared with conventional fracture propagation theory, more than 50% of hydraulic fractures in such reservoirs extend along bedding planes. Bedding planes at different positions affect fracture propagation, and differences in interlayer lithology lead to the initiation of fractures and bedding planes, forming three different types of fracture–bedding morphologies. After fracture creation, fractures may close rapidly once pumping stops. Supported fractures show significantly higher permeability than unsupported ones, making effective proppant placement in fracture networks during the proppant-laden stage a key factor in productivity enhancement. Research on proppant transport and placement in hydraulic fractures has been carried out through experimental and numerical simulation approaches. In experimental studies, KERN et al. (1958) first simulated proppant transport in single-wing fractures, analyzing the effect of particle size in water and guar-based fluids. BABCOCK et al. (1967) used transparent organic glass plates to study the influence of fracturing fluid and fracture geometry on proppant transport. In 2014, Li Jing improved experimental devices to simulate proppant transport in both single and complex fractures, studying the effects of pumping rate, fluid viscosity, and sand ratio. In 2023, Guo Tiankui et al. conducted large-scale visual simulations on proppant transport and placement in complex fractures, focusing on fracture geometry, particle size, and sand ratio. In numerical simulations, HU et al. (2018) investigated proppant transport and placement in single fractures under different pumping methods, while in 2024, Gong et al. used a CFD–DEM coupling method to study bridging and accumulation mechanisms in narrow fractures, considering fluid leak-off effects. In the same year, Liu Chunting et al. used CFD–DEM to simulate proppant transport in fractures, analyzing the influence of fracture geometry, fracturing fluid properties, and proppant properties. At present, few experimental or numerical studies have simulated proppant transport and placement in complex fractures under bedding-plane opening conditions. Most existing research focuses on complex fracture networks in shale, with limited discussion from the perspective of bedding-plane opening, leading to insufficient theoretical support for understanding its influence on proppant transport. Considering that in deep coal seam fracturing, developed bedding planes are prone to opening, allowing proppant-laden fluid to flow not only in hydraulic fractures but also along bedding planes, it is therefore essential to study proppant transport and placement in fracture–bedding systems. This is important for accurately predicting and controlling fracturing outcomes and for providing a theoretical basis for optimizing treatment design. Compared with experiments, numerical simulation offers advantages in constructing fracture geometries and reducing costs. Accordingly, this study employs numerical simulation to systematically investigate proppant transport and placement in fracture–bedding systems under bedding-plane opening conditions during deep coal seam hydraulic fracturing, aiming to provide a more reliable theoretical and technical basis for fracturing design in low-permeability reservoirs.

Geological Setting

The 530 study area of the Karamay Oilfield is located in the northwestern edge of the Junggar Basin, and the reservoirs of the Kexia Formation in this area are in the lower plate of the South Baikalintan Fracture and the Kew Fracture, and the top structure of the reservoirs is a monoclinic slope dipping from northwestern to south-eastern, which is subjected to the pulling effect of the anticlinal fracture. The dip of the formation near the northwest fault reaches 45°, while the dip of the formation in the southeast slope area is between 3° and 5°. The single sand layer is mainly subphase deposition in the fan delta plain, and mainly develops microphase deposition in the diversion channel and estuarine sand dam, with stable development and good continuity, the thickness of the sand layer ranges from 6.0m to 12.0m, with an average of 8.5m, and the effective thickness of the sand layer is 1.75m. The test area is located in the southeastern extension of the reservoir of the Keshia Formation in the 530 well area of the eighth district. The tectonic morphology of this area is simple, a southeast-dipping monocline, and internal faults are not developed (Figure 1).

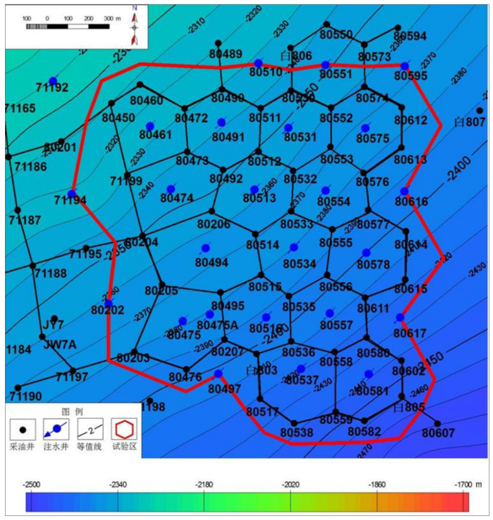
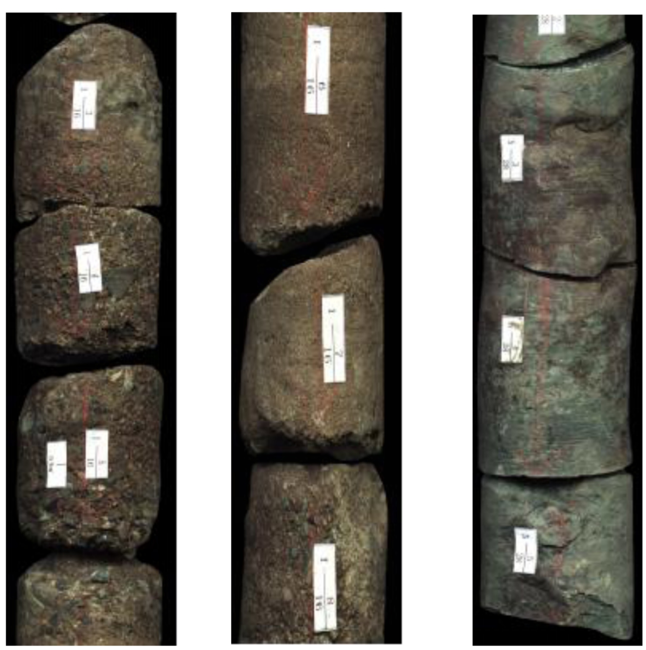


Figure 1—Bottom structure of study area

Lithological Assembly

The reservoir lithology in the test area mainly includes the following types: conglomerate, unequal granule conglomerate, sandy conglomerate, gravelly sandstone, conglomeratic sandstone, gritty sandstone, unequal granule sandstone, coarse sandstone, medium sandstone, fine sandstone, silty sandstone, argillaceous conglomerate, and argillaceous sandstone. Among these, sandy conglomerate and gravelly sandstone are the main reservoir rock types, followed by conglomeratic sandstone (Figure 2). The lithology generally coarsens downwards with dominant gravelly sandstone, sandy conglomerate, and gritty sandstone in the lower section, characterized by poor sorting and shapes ranging from sub-rounded to sub-angular; the middle section primarily consists of conglomeratic sandstone and medium to fine sandstone, also with poor sorting and predominantly sub-rounded shapes; the upper section is mainly composed of fine sandstone, silty mudstone, and mudstone, with poor sorting.



Grey fluorescent conglomerate Grey fluorescent gritty sandstone Grey-green mudstone

Figure 2—Core photographs from Bai 806 Well in the study area

Fracture Geometry Modeling

In 2015, Pandey et al. simulated the dimensions of primary hydraulic fractures in coal seams using multiple methods. Based on their results and field monitoring data, the length and height of a single-wing primary fracture were approximately 150 m and 25 m, respectively, with a fracture width of 5 mm. Due to the large geometric scale of fractures, numerical simulations face significant challenges, primarily reflected in the enormous number of computational grids, which require substantial computing resources and lead to long computation times. According to similarity principles, the Reynolds number of fluid flow in numerical simulation fractures and field fractures should be comparable. Given that fracture width is measured in millimeters while length and height are in meters, they are of different orders of magnitude. To ensure simulation accuracy while reducing computational complexity, the fracture length and height were scaled down by a factor of 100. Meanwhile, fracture width and proppant particle size were kept on the same order of magnitude to better simulate proppant transport behavior. In field operations, bedding-plane opening depends on formation conditions and treatment parameters; in simulations, the bedding size was set equal to the fracture length in both the fracture length direction and the direction perpendicular to the fracture. The fracture–bedding models were built under the following assumptions: (1) no fluid leak-off at fracture–bedding walls; (2) fracture–bedding extends toward the fracture tip; (3) fracture–bedding wall roughness is ignored; and (4) both fracture and bedding geometries are rectangular prisms. Here, F1B denotes a primary fracture with one bedding plane, F2B denotes a primary fracture with two bedding planes, and F3B denotes a primary fracture with three bedding planes, where bedding width is 2.5 mm and fracture width is 5 mm. Based on different bedding-opening scenarios, various three-dimensional fracture–bedding models were established and subsequently used to simulate proppant transport and placement.

Mathematical Formulation

Proppants are carried into fractures by the fracturing fluid, and their movement process is a typical solid–liquid two-phase flow. In computational fluid dynamics (CFD), multiphase flow models are generally classified into Euler–Euler and Euler–Lagrange models. The key difference lies in the treatment of the particle phase: in the Euler–Euler model, the particle phase is treated as a continuous phase, whereas in the Euler–Lagrange model, it is treated as a discrete phase. The Euler–Euler model offers advantages such as lower computational demand, shorter simulation time, and better convergence. Therefore, the Euler–Euler model was adopted to simulate proppant transport and placement within the three-dimensional fracture–bedding model. In the Euler–Euler flow model, the continuity equations for the fracturing fluid and proppants are expressed as follows:

$$\frac{\partial}{\partial t}\alpha_l\rho_l + \nabla\alpha_l\rho_l\vec{v_l} = 0 \tag{1}$$

$$\frac{\partial}{\partial t}\alpha_p\rho_p + \nabla\alpha_p\rho_p\vec{v_p} = 0 \tag{2}$$

where α_1 and α_p represent the volume fractions (%) of the fracturing fluid and proppants, respectively, with $\alpha_l + \alpha_p = 1$; v_1 and v_p denote the velocities of the fracturing fluid and proppants (m/s); and p_1 and p_p are the densities of the fracturing fluid and proppants (kg/m³), respectively.

In the Euler–Euler flow model, the momentum conservation equations for the fracturing fluid and proppants are expressed as:

$$\frac{\partial}{\partial t}\left(\alpha_l\rho_l\vec{v_l}\right) + \nabla\left(\alpha_l\rho_l\vec{v_l}\vec{v_l}\right) = -\alpha_l\nabla\rho_l + \nabla\cdot\overline{\overline{\tau_l}} + \alpha_l\rho_l g + \sum_{p=1}^{n}\left(\overrightarrow{R_{pl}} + m_{lp}\dot{\vec{v}}_{lp} - m_{pl}\vec{v}_{pl}\right) + \left(\overrightarrow{F_l} + \overrightarrow{F_{lift,l}} + \overrightarrow{F_{vm,l}} + \overrightarrow{F_{wl,l}} + \overrightarrow{F_{td,l}}\right) \tag{3}$$

$$\frac{\partial}{\partial t}\left(\alpha_p\rho_p\vec{v}_p\right) + \nabla\left(\alpha_p\rho_p\vec{v_p}\vec{v_p}\right) = -\alpha_p\nabla\rho_p + \nabla\cdot\overline{\overline{\tau_p}} + \alpha_p\rho_p g + \sum_{q=1}^{n}\left(\overrightarrow{R_{lp}} + m_{pl}\dot{\vec{v}}_{pl} - m_{lp}\vec{v}_{lp}\right) + \left(\overrightarrow{F_p} + \overrightarrow{F_{lift,p}} + \overrightarrow{F_{vm,p}} + \overrightarrow{F_{wl,p}} + \overrightarrow{F_{td,p}}\right) \tag{4}$$

Where p_1 and p_p are the partial pressures of the fracturing fluid and proppants, respectively (Pa); τl and τp are the stress–strain tensors of the fracturing fluid and proppant phases (N/m²); g is the gravitational acceleration (m/s²), typically taken as 9.81; $\dot{m}_{p1}$ land $\dot{m}_{1p}$ denote the mass transfer between the fracturing fluid and proppants (kg); F_1 and F_p are the external forces acting on the fracturing fluid and proppants (N); $F_{lift,1}$ and $F_{lift,p}$ are the lift forces acting on the fracturing fluid and proppants (N); $F_{w1,1}$ and $F_{w1,p}$ are the wall lubrication forces (N);$F_{vm,p}$ are the virtual mass forces (N); $F_{td,1}$, $F_{td,p}$ are the turbulent dispersion forces (N); and R_{1p} 、 R_{p1} represent the interaction forces between the fracturing fluid and proppants (N). In most cases, the magnitudes of the virtual mass and lift forces are negligible, and thus they are omitted from the calculation.

The stress–strain tensor is calculated as follows, using the fracturing fluid as an example.:

$$\overline{\overline{\tau_l}} = \alpha_l \mu_l \left(\nabla \vec{v_l} + \nabla \vec{v}_l^T\right) + \alpha_l \left(\gamma_l - \frac{2}{3}\mu_l\right) \nabla \cdot \vec{v_l} \overline{\overline{I}} \quad (5)$$

The interphase interaction force is calculated as follows:

$$\sum_{l=1}^{n} \overrightarrow{R_{lp}} = \sum_{l=1}^{n} \overrightarrow{k_{lp}} \left(\vec{v_l} + \vec{v_p}\right) \quad (6)$$

The turbulence model adopts the realizable k-ε model, with governing equations for turbulent kinetic energy(k):

$$\frac{\partial}{\partial t}(pk) + \frac{\partial}{\partial t}(pk\vec{v}) = \nabla\left[\left(\mu + \frac{\mu_t}{\sigma_k} \nabla k\right)\right] + G_k + G_b - \rho\varepsilon - Y_M + S_K \quad (7)$$

and dissipation rate:

$$\frac{\partial}{\partial t}\left(p_l \varepsilon\right) + \frac{\partial}{\partial t}(p\varepsilon\vec{v}) = \nabla\left[\left(\mu + \frac{\mu_t}{\sigma_\varepsilon} \nabla \varepsilon\right)\right] + \rho C_1 S_\varepsilon - \rho C_2 \frac{\varepsilon 2}{k + \sqrt{v\varepsilon}} + C_{1\varepsilon} \frac{\varepsilon}{k} C_{3c} G_b + S_\varepsilon \quad (8)$$

Where:

$$\theta = s\frac{k}{\varepsilon} \quad (9)$$

$$S = \sqrt{2 S_{ij} S_{ji}} \quad (10)$$

$$\mu_t \rho C_\mu \frac{k^2}{\varepsilon} \quad (11)$$

In these equations, G_k represents the turbulence kinetic energy generated by the mean velocity gradient (m²/s²), G_b represents the turbulence kinetic energy generated by buoyancy (m²/s²), and Y_M represents the contribution of fluctuating dilatation in compressible turbulence to the overall dissipation rate. C_2 、 $C_{1\varepsilon}$ and are constants with values of 1.9 and 1.44, respectively; σ_k and σ_ε are the turbulent Prandtl numbers with values of 1.0 and 1.2, respectively. The source terms S_k 、 S_ε and ε are set to zero.

Simulation Design

The transport and placement of proppants in hydraulic fractures are influenced by multiple factors, which can be broadly categorized into two groups: (1) geological factors, such as lamination aperture and the number of opened laminations; and (2) engineering factors, such as injection rate, fracturing fluid viscosity, sand ratio, and proppant particle size. This study investigates the effects of geological factors by analyzing how lamination aperture and the number of laminations influence proppant transport and placement. For engineering factors, the study examines the effects of varying injection rates (for a single stage with four clusters), fluid viscosity, sand ratio, and proppant size. To ensure that the simulated sand-laden fluid behavior is comparable to actual fracture conditions, the simulation uses the in-fracture flow velocity under field operating parameters as the standard. In practice, perforations are typically circular; however, for computational convenience, circular perforations are converted into square perforations, with the specific calculation as follows:

$$h = \frac{\pi D^2}{4w} \tag{12}$$

Where: h is the height of the square perforation (m); D is the diameter of the perforation (m); w is the fracture width (m). The injection rate corresponding to the flow velocity in the main fracture is calculated using the following formula:

$$v_f = \frac{Q_f}{A} \tag{13}$$

Where: V_f is the proppant-laden fluid velocity in the fracture (m/s); Q_f is the flow rate in a single fracture (m^3/s); A is the fracture cross-sectional area (m^2); Q is the injection rate (m^3/s). In numerical simulations, the inlet velocity of the fracture is calculated using the following formula:

$$v_m = \frac{v_f \times A}{nA_i}(1-\alpha) \tag{14}$$

Where: v_m is the inlet velocity of the fracture in the numerical simulation (m/s); n is the number of perforations; A_i is the inlet cross-sectional area (m^2); and α is the velocity attenuation coefficient, typically 0.2. Based on the field injection rate, the calculated inlet velocities for numerical simulations are shown in Table 1. In the simulations, the initial condition is set using the inlet velocity, with the inlet and outlet boundaries specified as velocity inlet and pressure outlet, respectively.

Table 1—Numerical simulation inlet flow rate calculation

Construction Rate/(m3/min)	Corresponding In-Fracture Velocity/(m/s)	Numerical Simulation Inlet Velocity/(m/s)
12	0.4	1.7
16	0.53	2.26
20	0.67	2.83

Based on the above analysis, Tables 2 and 3 summarize the simulation plans.

Table 2—Numerical Simulation Plan for Geological Factors

Scheme	Fracture Geometry	Bedding Opening /mm	Remarks
1	F1B	2.5	Effect of Bedding Opening
2	F1B	1	
3	F1B	0.5	
4	F1B	2.5	Effect of Number of Bedding Planes
5	F2B	2.5	
6	F3B	2.5	

Table 3—Numerical Simulation Plan for Engineering Factors

Scheme	Construction Displacement / (m³/min)	Fracturing Fluid Viscosity / mPa·s	Sand Ratio / %	Proppant Size / mesh
7	12	1	10	30
8	16	1	10	30
9	20	1	10	30
10	12	1	10	30
11	12	5	10	30
12	12	10	10	30
13	12	1	10	30
14	12	1	15	30
15	12	1	20	30
16	12	1	10	30
17	12	1	10	40
18	12	1	10	50

Result Analysis

Proppant-laden slurry enters the main hydraulic fracture, where the proppant (density 2650 kg/m³) settles at the fracture bottom due to its higher density compared to the fracturing fluid (1000 kg/m³), forming a sandbank. In contrast, within laminations, the proppant remains predominantly suspended and fills the lamination space. Proppant placement in the main fracture progresses through three stages: (1) bottom-settling stage (a), where the sandbank height reaches the lowest fracture entry point; (2) height-limited stage (b), where the sandbank height reaches the height from the fracture bottom to the lower lamination (HLB); and (3) sandbank growth stage (c, d), where the sandbank continues to expand with ongoing slurry injection. In laminations, proppants are mainly present in a suspended state.

Lamination Aperture

Under varying lamination aperture conditions, smaller apertures result in longer proppant transport distances and greater horizontal sandbank extent in the main hydraulic fracture, while proppant migration range within the laminations first increases and then decreases. During the height-limited stage, a smaller aperture leads to larger sandbank coverage in the main fracture but reduced proppant migration in the laminations. In the sandbank growth stage, when the aperture is 2.5 mm, sandbank height is constrained by the HLB; at 1 mm and 0.5 mm, height is unconstrained, though at 0.5 mm, the sandbank exceeds the HLB and causes sand bridging. Within laminations, proppants tend to distribute along the perimeter, but at 0.5 mm aperture, they cannot fully fill the lamination. As aperture decreases, entry resistance increases, favoring transport along the fracture length and promoting stable sandbank formation, while the higher fracturing fluid content in laminations accelerates proppant settling in the fracture. When the aperture is 0.5 mm, the resulting sandbank shape in the main fracture is distinct. Under constant operational parameters, there exists a minimum aperture at which sandbank height can exceed the HLB and stable proppant placement can be achieved, though excessively narrow apertures increase the risk of bridging. In practice, lamination aperture is closely related to pumping rate, and due to model limitations (treating proppants as a pseudo-fluid), simulated proppants can enter narrower laminations than in reality.

Lamination Counts

Under varying lamination counts, both proppant transport distance and stable sandbank length in the main hydraulic fracture decrease as the number of laminations increases. In laminations, suspended proppants exhibit an "apple-shaped" distribution with reduced presence near the fracture–lamination interface. In the upper laminations of F2B and F3B, proppant coverage expands with continued slurry injection, whereas in the middle lamination of F3B, distribution forms a "circular" pattern. During the sandbank growth stage, sandbank length in the main fracture continues to increase; in F1B, its geometry remains nearly unchanged, while in F2B and F3B, both sandbank length and height increase, though height is constrained by the HLB. In lower laminations, suspended proppants nearly fill the entire layer. Near the fracture–lamination interface, proppant concentration in F1B lower laminations is lower than in F2B and F3B due to the higher HLB position in F1B, which allows less proppant and more fracturing fluid to enter laminations during late-stage injection. Over time, suspended proppants migrate toward lamination boundaries, where concentrations are higher, indicating potential for far-field transport. In the middle lamination of F3B, proppants accumulate along the perimeter with higher concentrations at the boundary and a "Y-shaped" sand-free zone at the interface, while in the upper lamination of F3B, concentrations are nearly zero. Lower laminations closer to the fracture base promote more uniform sandbank formation, and both HLB height and lower lamination position significantly influence sandbank placement and proppant distribution.

Injection Rate

Under varying injection rates, higher rates result in longer stable sandbanks in the main hydraulic fracture and facilitate proppant transport to the far end. During the lamination height-limited stage, sandbank height is constrained by the HLB and does not increase further, while the proppant distribution in laminations expands, forming an "apple-shaped" pattern with higher concentrations at the tail and edges. In the sandbank growth stage, height remains unchanged, with only minor length increases; however, the horizontal distance from the sandbank crest to the slurry inlet, as well as the overlap with the lamination, both increase, indicating a larger sandbank placement area at higher rates. Proppants nearly fill the laminations, showing a "Y-shaped" high-concentration zone near the fracture–lamination intersection. Across all injection rates, sandbank geometry remains similar and height is mainly controlled by the HLB. In laminations, higher rates promote broader proppant migration toward the periphery, though stable sandbanks do not form, and late-stage injection yields uniform proppant distribution and concentration.

Fracturing Fluid Viscosities

Under varying fracturing fluid viscosities, the stable sandbank length in the main hydraulic fracture increases with viscosity. In the lamina, the proppant migration distance along the fracture length exceeds that along the width. Although higher viscosity enhances the slurry's proppant-carrying capacity and allows proppant to reach the far end of the main fracture, it does not effectively promote lateral migration within the lamina. Specifically, the migration distance along the fracture length remains greater than that along the width. During the lamina height-limiting stage, when viscosities are 1 mPa·s and 5 mPa·s, the sandbank height is limited by the HLB, with the 5 mPa·s case producing two distinct crests. At 10 mPa·s, the sandbank height fails to reach the HLB despite improved proppant transport to the far end of the main fracture. The lateral migration pattern in the lamina also changes with viscosity: at low viscosity, proppant spreads in an "arc" pattern, while at higher viscosity it advances linearly toward the lamina boundaries. At 10 mPa·s, the proppant nearly fills the entire lamina. During the sandbank expansion stage, viscosities of 1 mPa·s and 5 mPa·s lead to small lengthwise growth with minimal shape change, while at 10 mPa·s the sandbank height increases slightly but never reaches the lamina height. Within the lamina, viscosities of 1 mPa·s and 5 mPa·s result in near-complete filling, with suspended proppant near the fracture–lamina junction forming a "Y"-shaped distribution, and with higher concentrations observed at 1 mPa·s than at 5 mPa·s due to reduced settling at lower viscosity.

Sand Concentration Ratio

During the basal sedimentation stage, increasing the sand ratio reduces the stable sandbank length in the main fracture. Although the proppant concentration in the main fracture becomes higher, its migration distance decreases. This occurs because a higher sand ratio enhances particle–particle interactions, causing greater kinetic energy loss and resulting in proppant settling near the slurry inlet. During the lamina height-limiting stage, the sandbank height is constrained by the HLB and no longer increases, while the horizontal distance from the inlet to the sandbank crest decreases with increasing sand ratio. In the lamina, as the sand ratio increases, the distribution distance along the fracture length becomes shorter than that along the fracture width, and the overall proppant concentration increases.

Proppant Sizes

Under varying proppant sizes, the simulated distribution of proppant in the main hydraulic fracture and lamina is shown in Figure 10. During the basal sedimentation stage, smaller proppant sizes lead to increased stable sandbank length and longer migration distance in the main fracture, with noticeable discontinuities near the inlet. In the lamina, fine proppant (50 mesh) exhibits a longer distribution along the fracture length than along its width. During the lamina height-limiting stage, smaller particle sizes increase the horizontal distance from the inlet to the sandbank crest in the main fracture. For fine proppant (40 and 50 mesh), two distinct peaks form at the sandbank top, with the right peak developing earlier than the left—this is attributed to particle-size-dependent variations in the angle of repose. In the lamina, fine proppant nearly fills the entire space, whereas coarse proppant forms an "apple-shaped" distribution with greater nonuniformity as particle size increases. In the sandbank growth stage, for 30-mesh proppant, the stable sandbank shape in the main fracture remains largely unchanged. For 40-mesh proppant, the left peak gradually rises to the HLB, while for 50-mesh proppant, the left peak ceases to grow further, with the lamina becoming nearly fully packed. Near the fracture–lamina interface, proppant concentration decreases with smaller particle size, but the overall concentration distribution across the lamina remains relatively uniform.

Conclusions

This study investigates the transport and placement of proppant in hydraulic fractures and laminae within deep coal seams, focusing on fracture–lamina configurations distinct from shale fracture networks. Using a Euler–Euler approach, a numerical model was developed to simulate proppant migration under both geological and operational influences. The main conclusions are as follows: Proppant transport in fracture–lamina systems occurs in three stages: basal sedimentation, lamina height-limiting, and lamina filling. Lamina opening significantly affects the sandbank height in the main fracture, with a critical aperture at which the sandbank height exceeds the vertical distance from the main fracture base to the lower lamina (HLB). As lamina aperture decreases, proppant entry becomes more difficult, favoring longitudinal migration in the main fracture and sandbank formation, though excessively narrow laminae pose a sand plugging risk. With multiple laminae, sandbank height is constrained by HLB and does not exceed it; increasing the number of opened laminae also promotes fracturing fluid leak-off. Higher injection rates increase sandbank coverage and enhance proppant migration toward lamina boundaries. High-viscosity fracturing fluids limit sandbank height below HLB and hinder lateral proppant transport. Increasing sand concentration improves uniform placement in the lower main fracture but still constrained by HLB; it also raises proppant concentration in laminae but reduces long-range transport. Smaller proppant sizes extend the horizontal reach of the sandbank top, yield more uniform lamina placement, and facilitate radial migration. Multiple "peaks" may form in sandbanks depending on particle size, with smaller particles giving more uniform lamina distribution. Under lamina conditions, reducing lamina aperture or increasing lamina count decreases the characteristic length and area of stable sandbanks in the main fracture compared to a single vertical fracture, with reductions of 1.33%–26.67% in length and 33.33%–50.37% in area. Under varying operational conditions, reductions range from 0.67%–34.67% in length and 36.88%–49.06% in area.

Acknowledgements

The authors would like to thank (CNPC) China National Petroleum Corporation for the permission of publishing this work. The financial support for project research from National Key Research and Development Program: Key technical standard system and demonstration for CO2 storage in large-scale oil and gas reservoirs(2023YFF0614104), National Science and Technology Major Project: Research on Collaborative Optimization and Control Technology of CO2 Oil Displacement and Storage(2024ZD1406605), Science and Technology Special Project of Xinjiang Oilfield Company: Engineering Research and Field Test of CO2 Flooding Geological Reservoir in the Junggar Basin(2024XJZX0801).

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